

CNC Milling Machine — Operation & Reference Manual

Equipment Manual · Machine Type: CNC Milling

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Owner	Manufacturing Engineering
Applies To	CNC Line — Plant A (Pune), Plant B (Chennai); units CNC_Milling_1 through CNC_Milling_20

1. Overview

This manual covers the standard 3-axis CNC milling centers deployed on the CNC Line at Plant A and Plant B. Each unit performs precision milling of metal components fed from the upstream stock cutting stage and feeds finished parts to the Assembly Line for downstream fitment.

2. Machine Specifications

Parameter	Specification
Spindle speed range	500 – 12,000 RPM
Ideal cycle time	1.0 – 1.4 min/part (per fact_production ideal_cycle_time baseline)
Rated planned runtime	480 min/shift
Tool changer capacity	20-station automatic tool changer
Coolant system	Flood coolant, 40 L reservoir
Control system	Fanuc-compatible CNC controller

3. Startup Procedure

- 1 Confirm hydraulic clamp pressure gauge reads within 55–65 bar before energizing spindle.
- 2 Power on control cabinet; wait for controller self-test to complete (approx. 45 sec).
- 3 Home all axes (X, Y, Z) using the reference return cycle.
- 4 Load the correct part program and verify tool offsets against the Calibration_SOP record for the shift.
- 5 Run first article at reduced feed override (50%) and inspect against drawing tolerance before full-rate production.

4. Shutdown Procedure

- Return spindle and axes to home position.
- Retract coolant nozzle and stop coolant pump.
- Clear swarf/chips from the enclosure and coolant tray.
- Log shift production counts and any downtime codes in the shift handover sheet.

5. Common Downtime Codes on This Machine Type

Code	Description	Severity	Reference SOP
DT-01	Hydraulic System Failure	Major	Hydraulic_Maintenance_SOP
DT-03	Tooling / Die Changeover Delay	Moderate	Changeover_SOP
DT-04	Sensor / Limit Switch Fault	Minor	Electrical_Troubleshooting_SOP
DT-06	Calibration Drift	Moderate	Calibration_SOP
DT-08	Spindle / Main Bearing Failure	Major	Bearing_Replacement_SOP

■ **Never open the enclosure door while the spindle is rotating. Lockout-tagout per Safety_Guidelines before any tool-change troubleshooting.**